

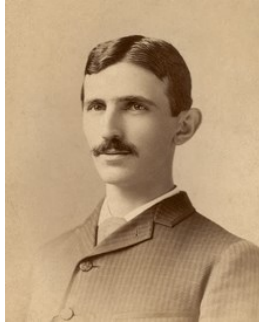
5G E2E QoS (Quality of Service)

Created: 2024_0324

Updated: 2025_0511

By JMA

Resource



**I don't care that they stole my idea . .
I care that they don't have any of their own**

~ Nikola Tesla ~

How to learn 5G QoS

Total: 1H + 40min



2024

“5G QoS” found in Youtube --1



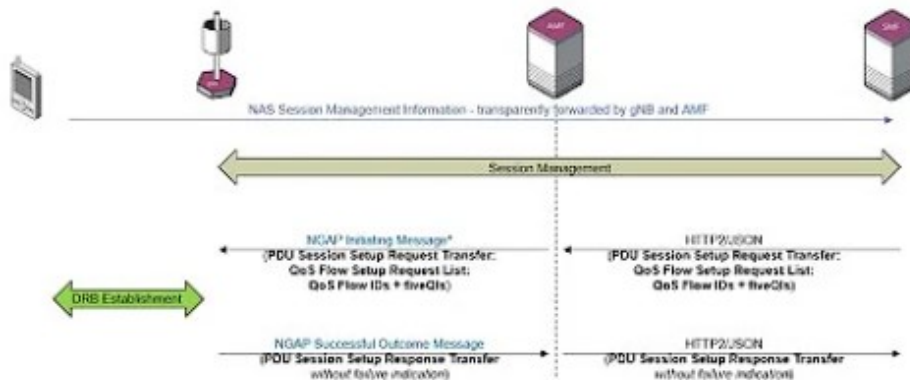
Quality of Service in 5G
5G Mobile Communications · 播放清單

Introduction to Quality of Service in 5G (1/12) · 1:26
PDU Session and QoS Flows (2/12) · 3:53
查看完整播放清單

<https://youtu.be/KS4bOW3J8m4?si=tl5Res-uFehBGf9v>

12 clips

QoS Flow Setup: gNB - AMF - SMF



QoS Flow Establishments in 5G RAN and Core

觀看次數：2861次 · 2 年前



Ralf Kreher

In this video I explain the fundamentals of establishing 5G QoS Flows while highlighting the difference between 4G and 5G user ...

1 clips

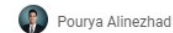
https://youtu.be/fnii_fA1W4s?si=5rhAa3jUjY8QEXyt

“5G QoS” found in Youtube --2



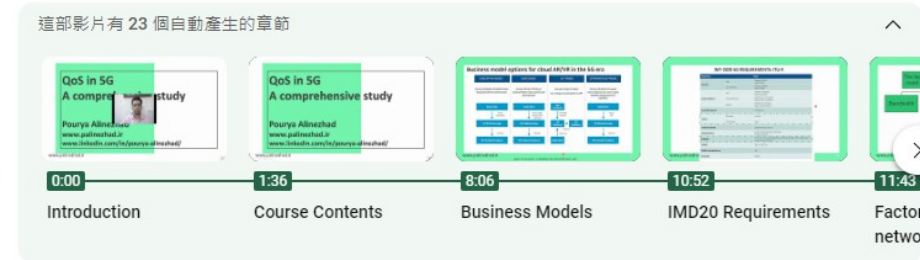
Qos in 5G (Quality of service in 5G) - Part 1

觀看次數：5492次 · 3 年前



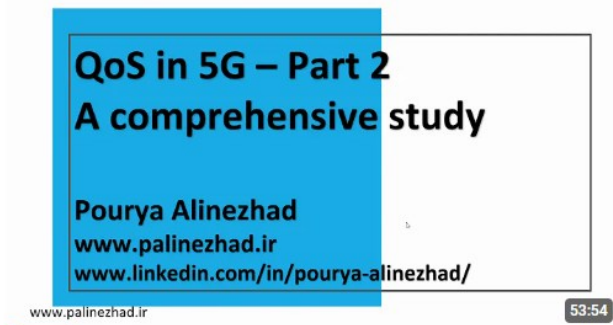
Pourya Alinezhad

This is a video on #5G #QoS and I uploaded it in two parts. The first part a brief review on 4G Qos and complete 5G QoS ...



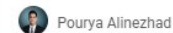
https://youtu.be/zERfUZ5UQ8Y?si=bWUN_8J1eoW08MPB

1 clips



Qos in 5G (Quality of service in 5G) - Part 2

觀看次數：2066次 · 3 年前



Pourya Alinezhad

This video on #5G #QoS part 2 includes QoS provisioning, enforcement and mapping, SDAP layer, Qos in 5G in NB-IoT, Network ...

字幕



1 clips

https://youtu.be/_TuC4BfOBOU?si=QLA83Eih3xT5UkIA

“5G QoS” found in Youtube --3



5G Course - 5G QoS Flow / SDAP layer and Reflective QoS

觀看次數：489次 · 1 年前



5G Understanding

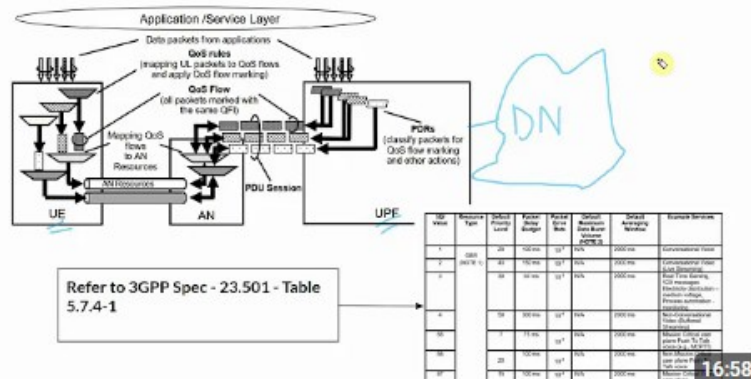
In this episode I tell you what is 5G QoS Flow, SDAP layer and why in 5G there is a new Reflective QoS. Check my Udemmy ...

1 clips

<https://youtu.be/XTEQDhmy8SY?si=H41JBR5mSew2R0yE>

SDAP (Service Data Adaptation Protocol)

5G QoS



QoS (Quality of Service) in 5G

觀看次數：9947次 · 3 年前



Telco Bytes

QoS in 5G is an integral part given the various 5G use cases. In this video we will discuss the key implementation details re 5G ...

Qos in 5g | Background Information | Implementation of Qos and 4g | Quality of Service Class Indicators |... 8 個段落

1 clips

<https://youtu.be/GUNQjwjEwQM?si=J5WmMfoxtglb8-Lf>

“5G QoS” found in WeChat --1

5G网络中业务数据的QOS机制

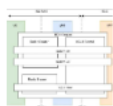


5g (nr) 延用了前两代 (3g, 4g) 移动技术, 控制面与用户面数据分离的机制; 这种机制包括: qos策略管理、信令和准入控制; 而qos服务策略管理涉及定义、级别和类型配置以及...

和老康一起学5G 3個月前

https://mp.weixin.qq.com/s/y8n71UT_QkNemGIfpTq3gQ

5G NR QOS简介

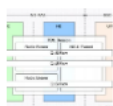


在4g中, qos是基于承载 (bearers), 而在5g中, qos是基于flows。在4g中, 使用eps承载标识符 (eps bearer id) 来区分不同的qos, 而在5g中我们使用qfi。注意: 4g qos应用在eps...

信息技术圈 2個月前

https://mp.weixin.qq.com/s/3fppz-mq_8LZZ_7IJlPhw

5G QOS 体系

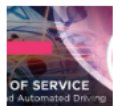


5g sa qos架构主要特点: 基于qos flow进行端到端的qos管理; ng-u呈现pdu session/qos flow, 不是e-rab; 每个pdu session内存在一条缺省qos flow及对应缺省qos rule贯穿整个生...

长沙臻杰科技 2022-4-24

<https://mp.weixin.qq.com/s/4OTTc257OBOe4EuHgFjfjw>

5G(NR)中QoS如何实施(一)?

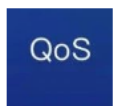


5g (nr) 无线系统中qos (服务质量) 是确保为不同类型服务和提供可靠且可预测性能的关键因素。系统中qos的实施涉及各网元机制和流程, 以根据特定服务对数据流进行优先级排序...

和老康一起学5G 4個月前

<https://mp.weixin.qq.com/s/UfWBaP2J2Amr1Wt-ESbK8w>

5G QoS



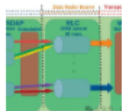
在4g中, qos是基于承载 (bearers), 而在5g中, qos是基于flows。在4g中, 使用eps承载标识符 (eps bearer id) 来区分不同的qos, 而在5g中我们使用qfi。4g qos应用在eps承载上。

我想我思 2個月前

<https://mp.weixin.qq.com/s/mcv8PNw4FyYqyrHsciQLIA>

“5G QoS” found in WeChat --2

QoS在5G(NR)系统中是如何应用的?



5g (nr) 无线系统中qos (服务质量) 是确保不同类型服务和应用提供可靠且可预测性能的关键。5g系统中qos涉及各种网元机制和程序, 根据特定服务要求对数据流进行优先级排序和...

和老康一起学5G 2 個月前

<https://mp.weixin.qq.com/s/9jfV31NQ2fwfUC6l5ivAfA>

5G如何保障服务质量?



qos作为5g的一大亮点功能, 所能做到的还远远不止于此。如果你的业务也对服务质量有很高的要求, 想要通过qos保障来提高服务的可用性和客户体验, 甚至拓展新的应用场景, 欢迎...

阿里云开发者 2021-4-1

<https://mp.weixin.qq.com/s/PsDtUlu6LtgbChEeGOOE0Q>

5G系统QoS



我想我思 4 個月前

Good



<https://mp.weixin.qq.com/s/SLsT878qan6qXwQiwELR6g>

在Flow层面, 5G是如何进行QoS控制的?



在5g网络中, qos控制分为三个层次, 包括业务数据流级别 (service data flow level)、qos flow级别和承载级别 (bearer level)。在业务数据流级别 (sdf level), qos控制适用于ip...

深圳市雅诺讯科... 8 個月前

<https://mp.weixin.qq.com/s/SBAafJQc6k9xHD0pOdgfPQ>

5G 新QoS流处理



iii. 服务质量: 如果下行分组是通过某个drb接收的, 并且考虑到drb表示总qos (即, 延迟、gbr、丢失等), 则没有qos流特定qos参数来 (预) 配置ue。将需要向ran通知qos profile (...)

余网优化 11 個月前

<https://mp.weixin.qq.com/s/WirJqUwUT6b0j0GHZ9t5iQ>

A Good TextBook



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A Good TextBook

Here is the professional English translation of your sections with source

1. 5G QoS模型的確定

2. 點對點的QoS控制

2.1 點對點的QoS控制之想法介紹

2.2 PCC規則的確定

2.3 QoS流的產生和設定

2.4 UE端使用的QoS規則

3. QoS參數

3.1 5QI以及對應的QoS特徵

3.2 ARP

3.3 串流速率控制參數

4. 反向映射QoS

4.1 為什麼引入反向映射QoS

4.2 反向映射QoS的控制機制

5. QoS通知控制

5.1 QoS通知控制介紹

5.2 候選QoS設定的引入

6. 結論

1. Determination of the 5G QoS Model

2. End-to-End QoS Control

2.1 Conceptual Introduction to End-to-End QoS Control

2.2 Establishment of PCC Rules

2.3 Generation and Configuration of QoS Flows

2.4 QoS Rules Applied at the UE Side

3. QoS Parameters

3.1 5QI and Corresponding QoS Characteristics

3.2 Allocation and Retention Priority (ARP)

3.3 Traffic Rate Control Parameters

4. Reflective QoS Mapping

4.1 Rationale for Introducing Reflective QoS

4.2 Control Mechanism of Reflective QoS

5. QoS Notification Control

5.1 Introduction to QoS Notification Control

5.2 Introduction to Candidate QoS Configuration

6. Conclusion

What We Learn here

1	Youtube has many good materials	
2	5G QoS is not a general topic in 5G books	
3	Learning Chinese is a plus to get more resource for 3G, 4G, 5G and 6G	

2025

DeepSeek brought cheap AI to physical life

DeepSeek leaves US AI firms racing to understand its success

By Kenrick Cai, Anna Tong and Jeffrey Dastin

January 30, 2025 7:00 AM GMT+8 · Updated 3 months ago



SAN FRANCISCO, Jan 29 - Developers at leading U.S. AI firms are praising the DeepSeek AI models that have leapt into prominence while also trying to poke holes in the notion that their multi-billion dollar technology has been bested by a Chinese newcomer's low-cost alternative.

<https://www.reuters.com/technology/artificial-intelligence/american-ai-firms-try-poke-holes-disruptive-deepseek-2025-01-28/>

LiNE on 2025/1/27

1月27日(一)

已讀 7

===== test for AI (GPT vs DeepSeek)

已讀 7
上午 12:55

Q: 畫出5G SA E2E call flow for registration procedure



已讀 7

👉 by ChatGPT

已讀 7
上午 12:58



已讀 7
上午 12:58

👉 by DeepSeek

已讀 7
上午 12:59

=====

已讀 6
上午 9:28

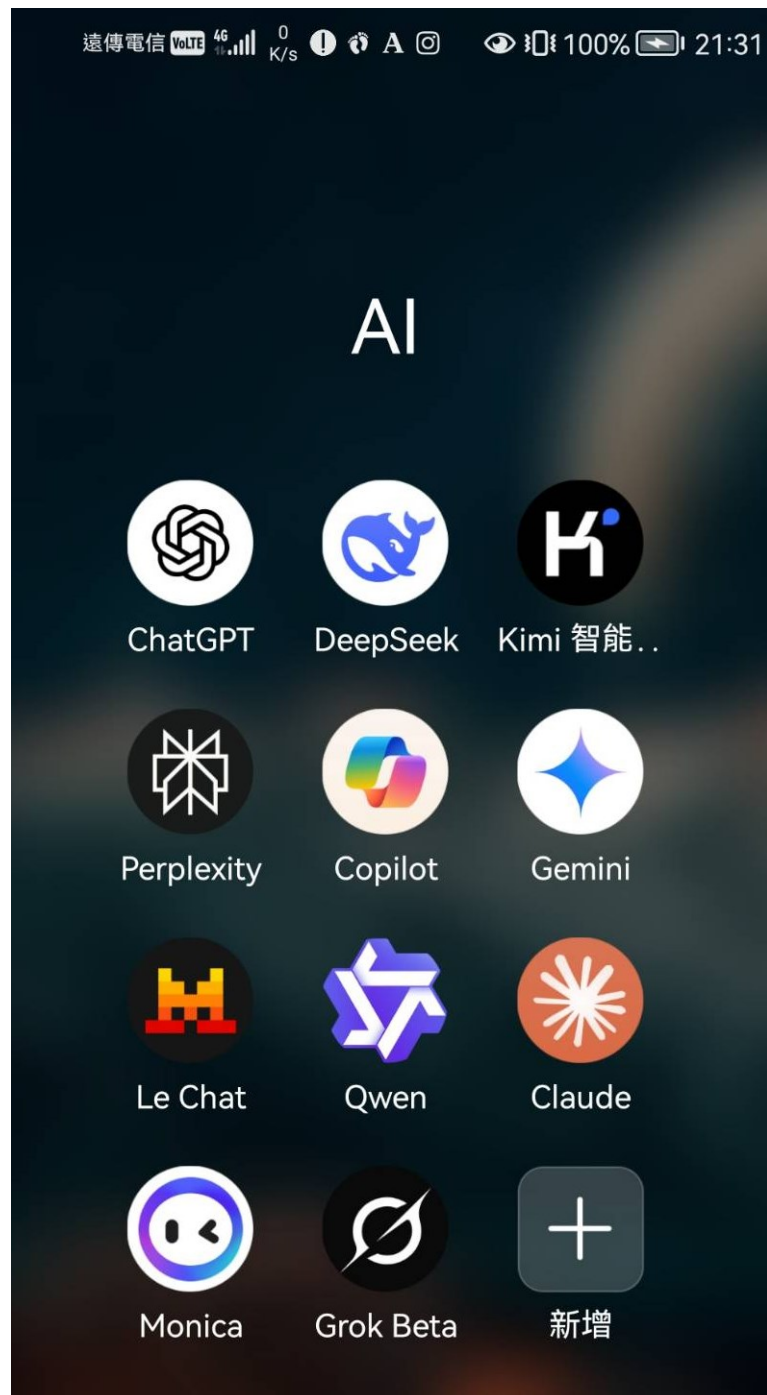
For 5G, it seems DeepSeek is better than GPT.

已讀 6
下午 1:47

Actually, after some tests, DeepSeek is much better than GPT.

已讀 6
下午 1:48

For 5G SA, GPT is just knowing nothing.



My AI Fellows

Here We Go

- Q: I am a teacher in university. Please provide a guideline in details for 5G End-to-end QoS. I will use it to teach Master's students.
- Q: I am a teacher in university. Please provide a guideline in details for QoS in WIFI. I will use it to teach Master's students.

Got it in just 10min

Optimizing Wi-Fi Quality of Service for Master's Students ▾

+

J

I am a teacher in university. Please provide a guideline in details for QoS in WIFI. I will use it to teach Master's students.

Quality of Service (QoS) in Wi-Fi Networks: A Comprehensive Guide

Document

I'd be happy to create a comprehensive teaching guide on Quality of Service (QoS) in Wi-Fi networks for your Master's students. This is an important topic for advanced networking courses.

Quality of Service (QoS) in Wi-Fi Networks: A Comprehensive Guide

Document

I've created a comprehensive teaching guide on Quality of Service (QoS) in Wi-Fi networks that's tailored for Master's level students. Here's what the guide includes:

Overview of Contents

The guide is structured with 16 detailed sections covering:

- 1. Introduction to QoS in Wi-Fi** - Fundamental concepts and importance
- 2. Theoretical Foundations** - Traffic requirements, QoS models, and mathematical

Reply to Claude...

+

Claude 3.7 Sonnet ▾

↑

Quality of Service (QoS) in Wi-Fi Networks

A Comprehensive Guide for Master's Level Instruction

Table of Contents

- [1. Introduction to QoS in Wi-Fi](#)
- [2. Theoretical Foundations of QoS](#)
- [3. IEEE 802.11 Standards and QoS](#)
- [4. Wi-Fi Multimedia \(WMM\) and Enhanced Distributed Channel Access \(EDCA\)](#)
- [5. QoS Parameters and Metrics](#)
- [6. QoS Implementation Techniques](#)
- [7. Traffic Classification and Prioritization](#)
- [8. QoS in Enterprise Wi-Fi Networks](#)
- [9. QoS in Modern Wi-Fi Standards \(Wi-Fi 6/6E/7\)](#)
- [10. Measurement and Analysis of QoS](#)
- [11. Challenges and Limitations](#)
- [12. Case Studies and Practical Applications](#)
- [13. Future Directions in Wi-Fi QoS](#)
- [14. Laboratory Exercises](#)
- [15. Discussion Topics and Research Questions](#)
- [16. References](#)

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How to use AI to learn 5G QoS

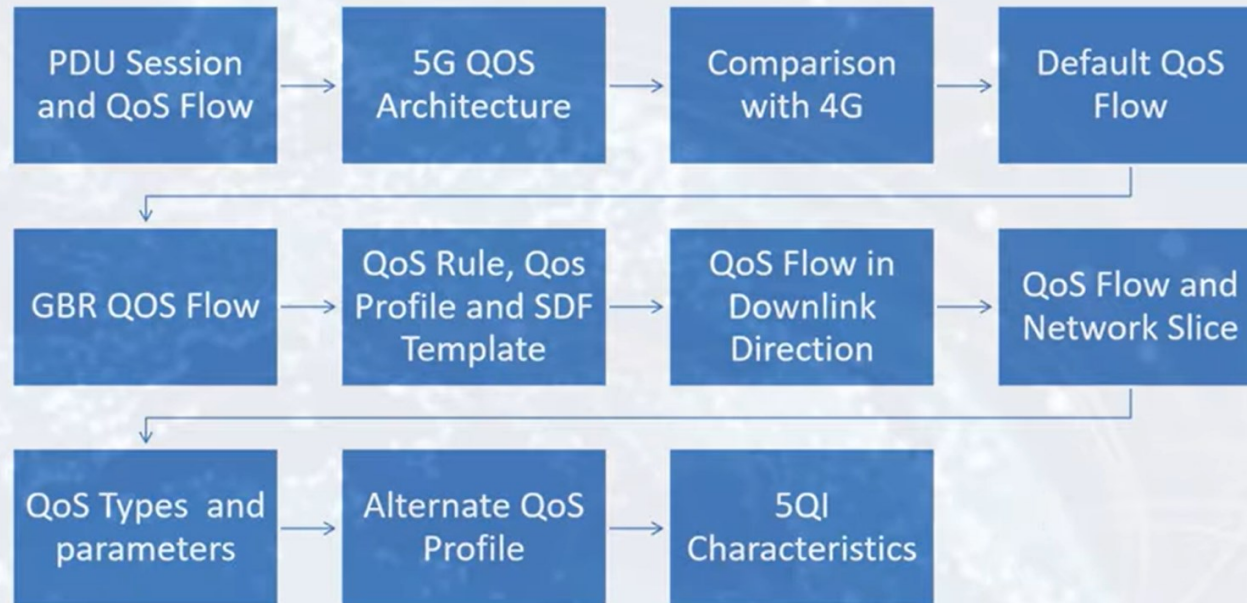
40min

What We Learn here

1	We can use free AI to speed up in learning curve	
2	Which AI is best tool for 5G QoS ?	
3		

Important pages

Content



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Important pages

<https://youtu.be/KS4bOW3J8m4?si=tl5Res-uFehBGf9v>

PDU Session and QoS Flows

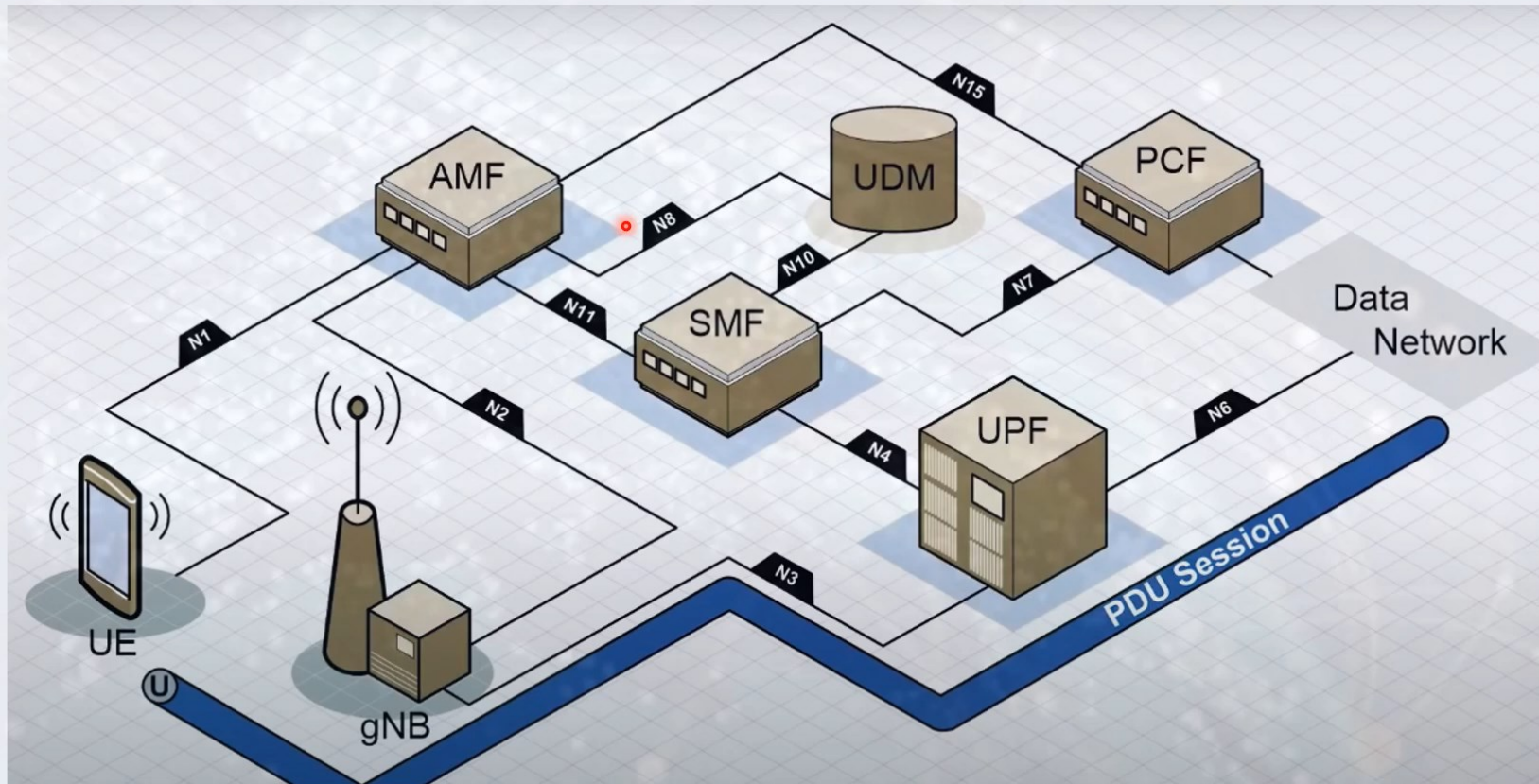


Image Source : <https://www.youtube.com/watch?v=YVoCpqsPwmQ&t=742s>

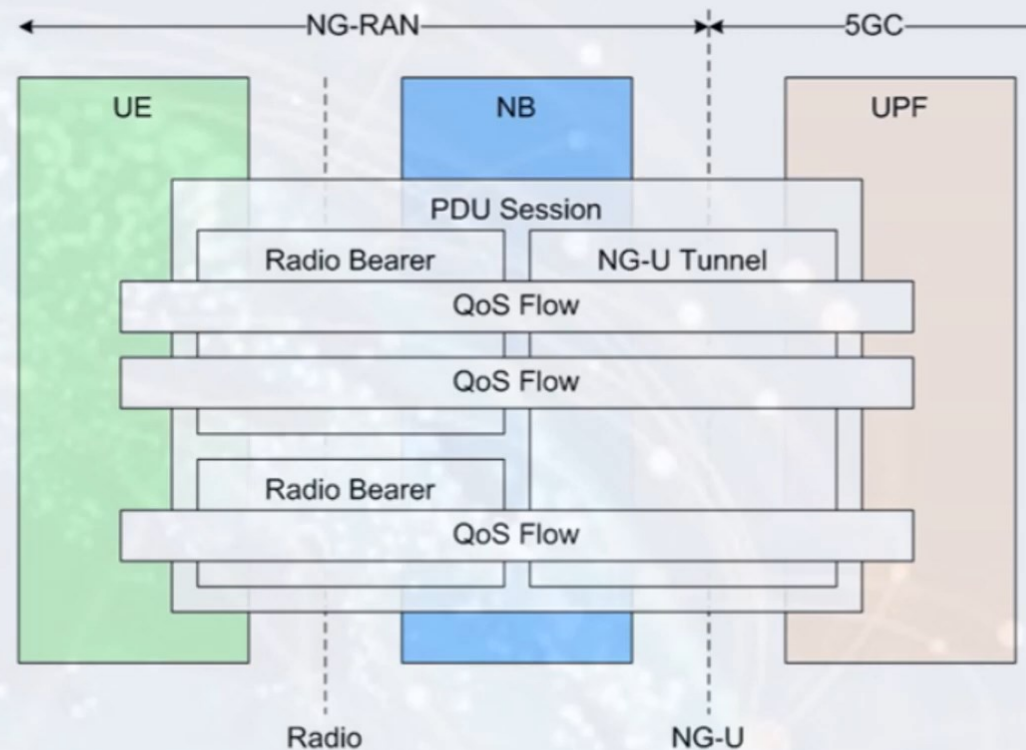
Important pages

<https://youtu.be/KS4bOW3J8m4?si=tl5Res-uFehBGf9v>

5G QoS Architecture

Types of QoS Flow

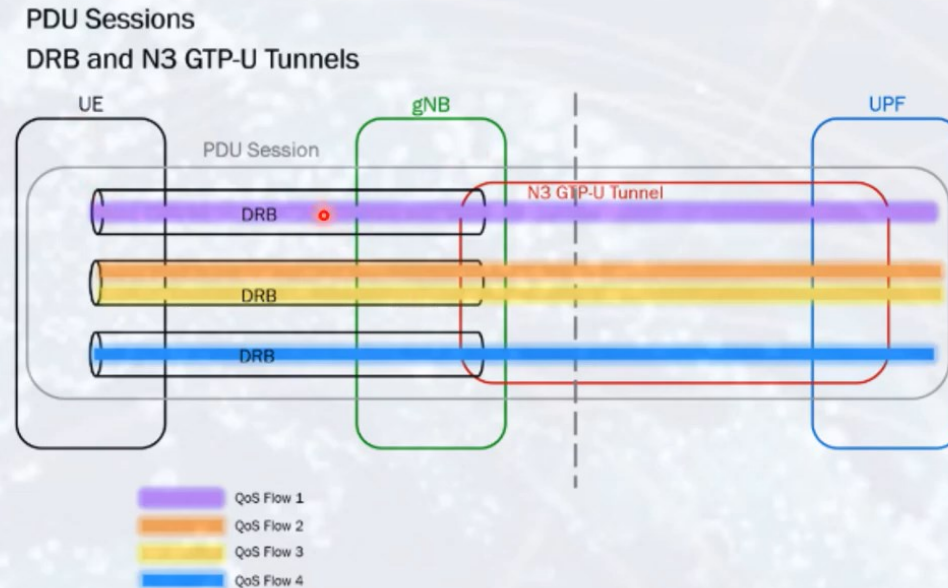
- QoS Flow with GBR
- QoS Flow with NGBR
- Delay Critical GBR
- Reflective QoS



Important pages

<https://youtu.be/KS4bOW3J8m4?si=tl5Res-uFehBGf9v>

5G QoS Architecture



A PDU session may contain multiple QoS flows and several DRBs but only a single N3 GTP-U tunnel. A DRB may transport one or more QoS flows.

DRB : Dedicated Radio Bearer
GTP : GPRS Tunnel Protocol

5G QoS Architecture

<https://clcnetwork.wordpress.com/2020/04/19/5g-core-pdu-session-and-qos-part-2/>

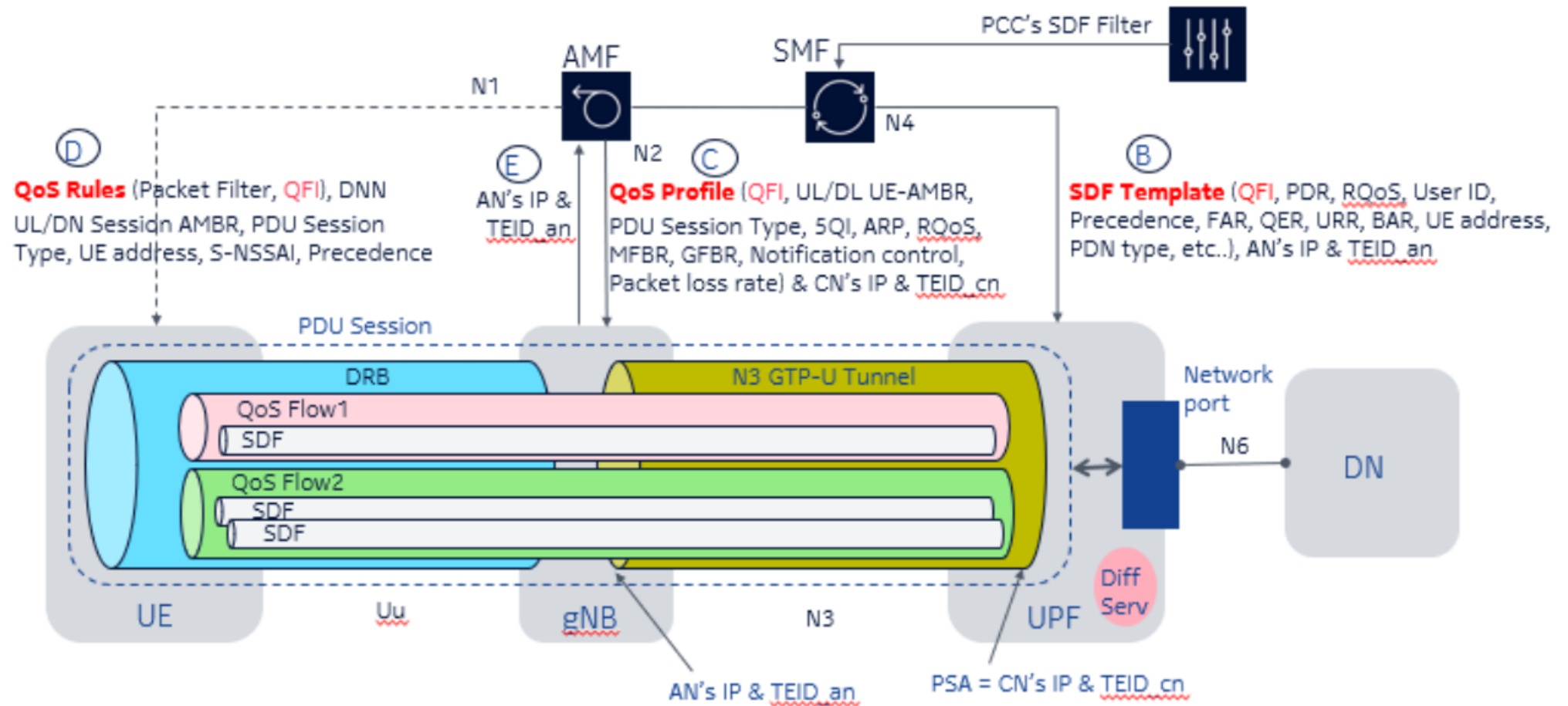


Figure 1 – SDF Template, QoS Profile and QoS Rule for UPF, gNB and UE respectively

5G QoS flows -- DL

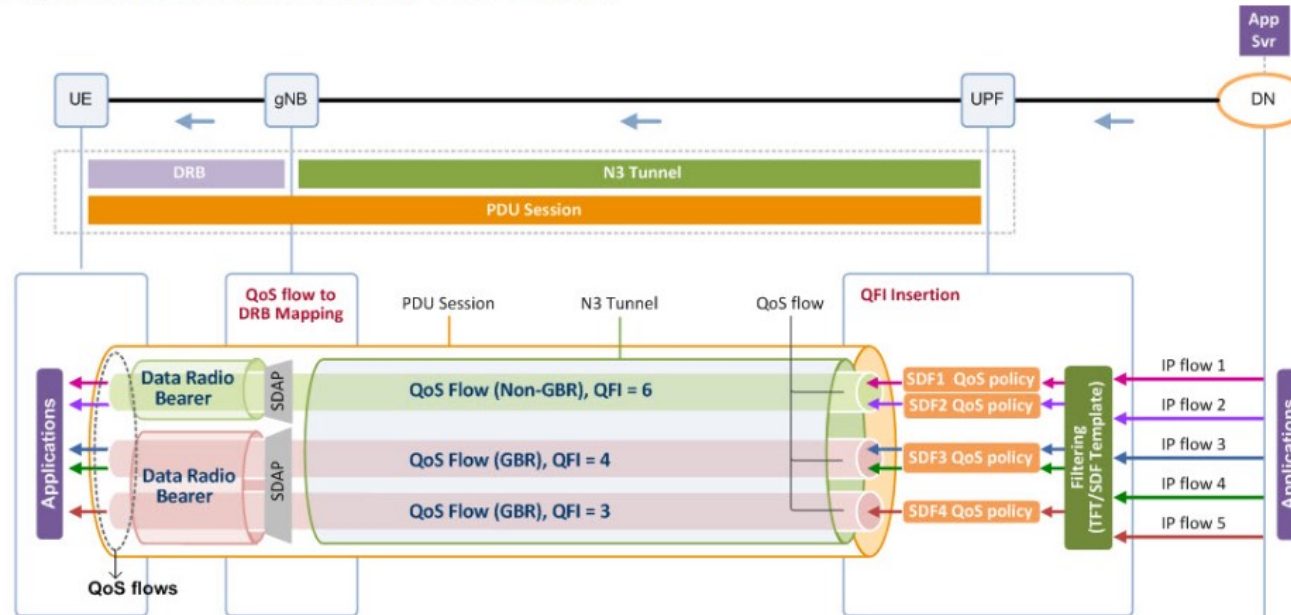
<https://netmanias.com/en/post/oneshot/14105/5g/5g-qos>

NETMANIAS ONE-SHOT

5G QoS - Downlink

February 7, 2019

The QoS differentiation within a PDU session



5QI : 5G QoS Identifier
 ARP : Allocation and Retention Priority
 GFBR : Guaranteed Flow Bit Rate
 MFBR : Maximum Flow Bit Rate
 PDB : Packet Delay Budget
 PER : Packet Error Rate
 QFI : QoS Flow Identifier
 RQA : Reflective QoS Attribute

QoS Flow type	QoS Flow parameters
GBR flow	Non-GBR flow
	5QI
	ARP
	RQA
	GFBR
	MFBR
	Notification Control
	Maximum Packet Loss Rate

5QI

Resource Type*
 Default Priority Level
 PDB
 PER
 Default Maximum Data Burst Volume
 Default Averaging Window

* GBR, non-GBR or delay critical GBR

5QI Characteristics

- Example of GBR

5G QoS Identifier | 5QI Table

Following table mentions 5QI values and their corresponding QoS characteristics mapping.

5QI Value	Resource Type	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
1	GBR (guaranteed flow bit rate)	20	100 ms	10^{-2}	N/A	2000 ms	Conversational Voice
2	GBR	40	150 ms	10^{-3}	N/A	2000 ms	Conversational Video (Live Streaming)
3	GBR	30	50 ms	10^{-3}	N/A	2000 ms	Real time gaming, V2X messages
4	GBR	50	300 ms	10^{-6}	N/A	2000 ms	Non-conversational Video (Buffered Streaming)
65	GBR	7	75 ms	10^{-2}	N/A	2000 ms	Mission Critical user plane Push to Talk voice (e.g. MCPTT)
66	GBR	20	100 ms	10^{-2}	N/A	2000 ms	Non-Mission Critical user plane Push to Talk Voice
67	GBR	15	100 ms	10^{-3}	N/A	2000 ms	Mission Critical Video user plane

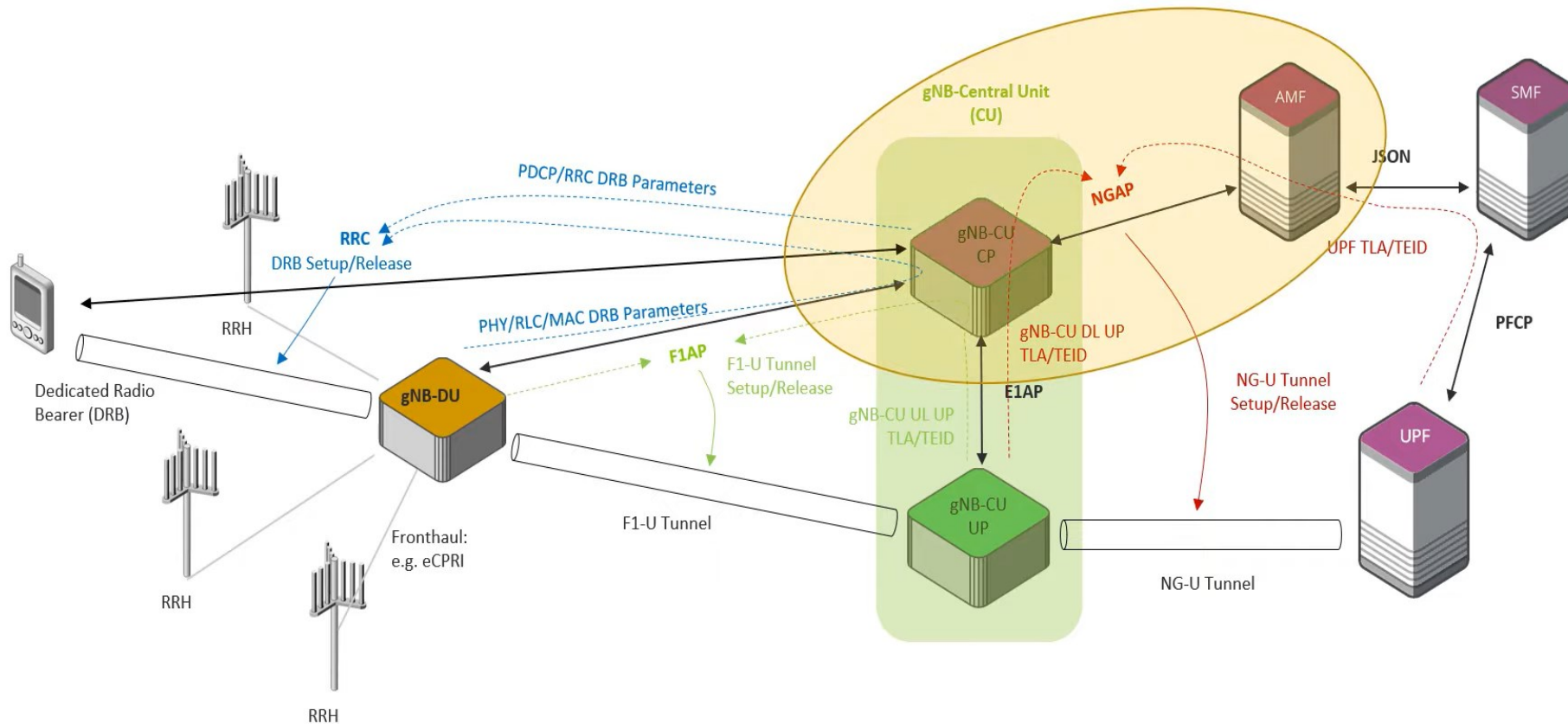
5QI Characteristics

- Example of NGBR

74	GBR	56	500 ms	10^{-4}	N/A	2000 ms	"Live" Uplink Streaming
76	GBR	56	500 ms	10^{-4}	N/A	2000 ms	"Live" Uplink Streaming
5	Non-GBR	10	100 ms	10^{-6}	N/A	N/A	IMS Signalling
6	Non-GBR	60	300 ms	10^{-6}	N/A	N/A	Video (Buffered Streaming) TCP-based (e.g. www, e-mail, chat, FTP, p2p file sharing, progressive video etc.)
7	Non-GBR	70	100 ms	10^{-3}	N/A	N/A	Voice, Video (Live Streaming), Interactive Gaming
8	Non-GBR	80	300 ms	10^{-6}	N/A	N/A	Video (Buffered Streaming) TCP-based (e.g. www, e-mail, chat, FTP, p2p file sharing, progressive video etc.)
9	Non-GBR	90	-As above-	-As above-	-As above-	-As above-	-As above-
69	Non-GBR	5	60 ms	10^{-6}	N/A	N/A	Mission Critical Delay Sensitive Signalling (e.g. MC-PTT Signalling)
70	Non-GBR	55	200 ms	10^{-6}	N/A	N/A	Mission Critical Data (e.g. example services are the same as 5QI 6/8/9)

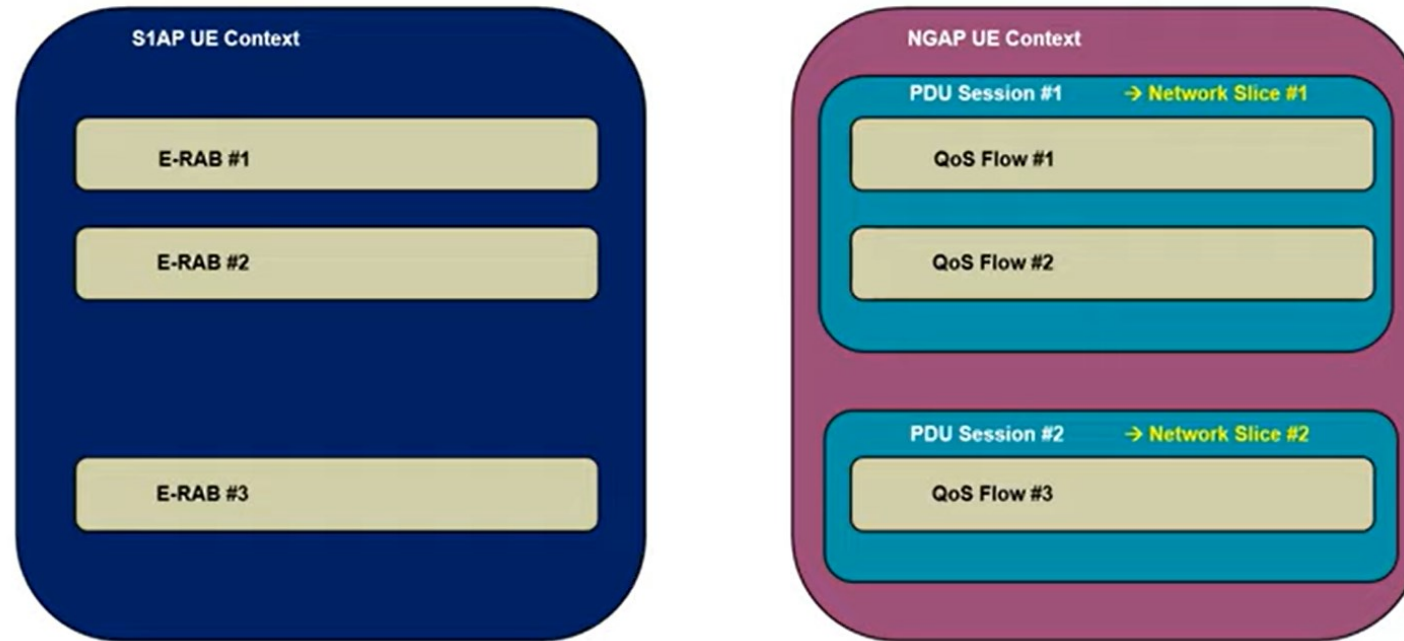
Important pages

Protocols Involved in Establishing UP Transport



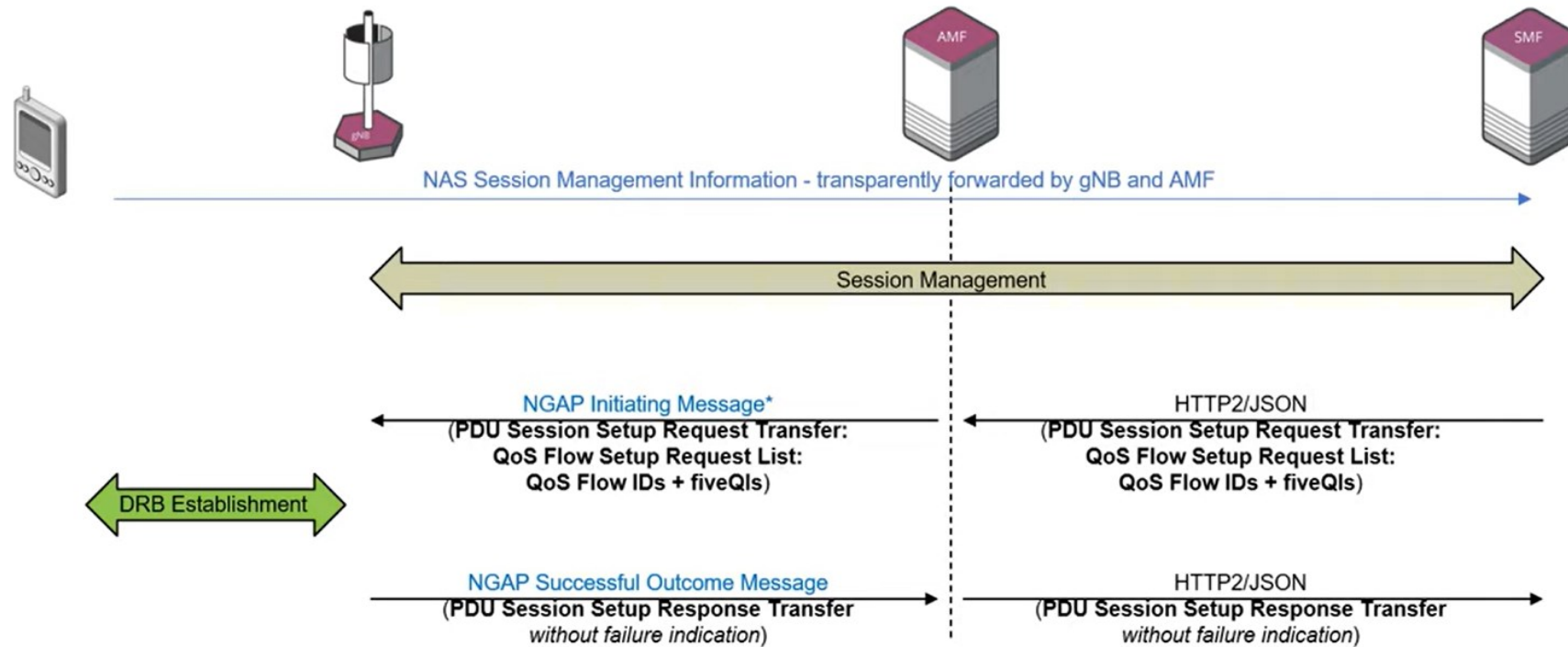
Important pages

NGAP UE Context vs. S1AP UE Context



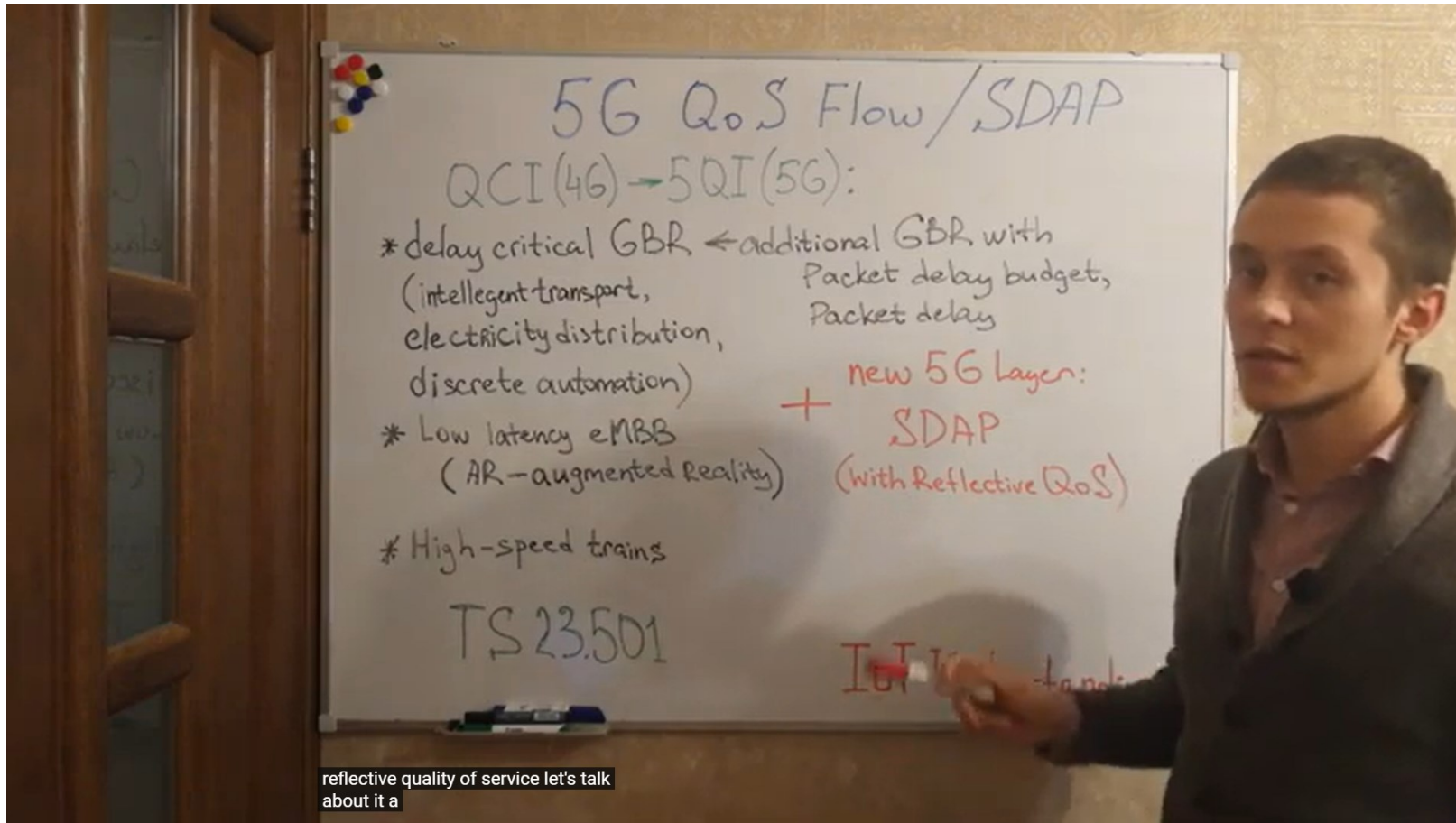
Important pages

QoS Flow Setup: gNB - AMF - SMF



* This can be:
NGAP Initial Context Setup Request
NGAP PDU Session Resource Setup Request
NGAP PDU Session Resource Modify Request
NGAP Handover Request (with different success/failure Response message)

Important pages

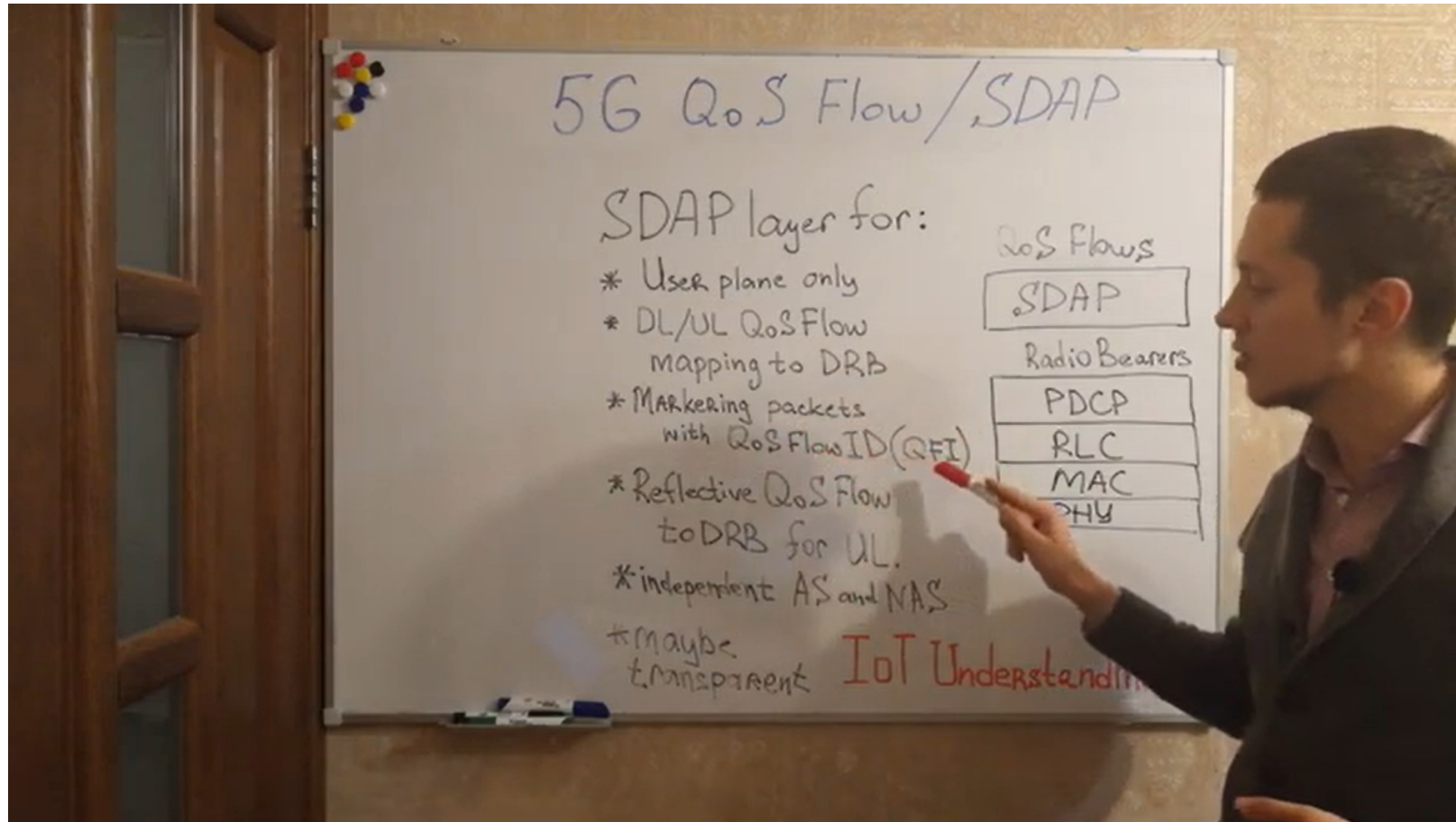


The whiteboard contains the following handwritten text:

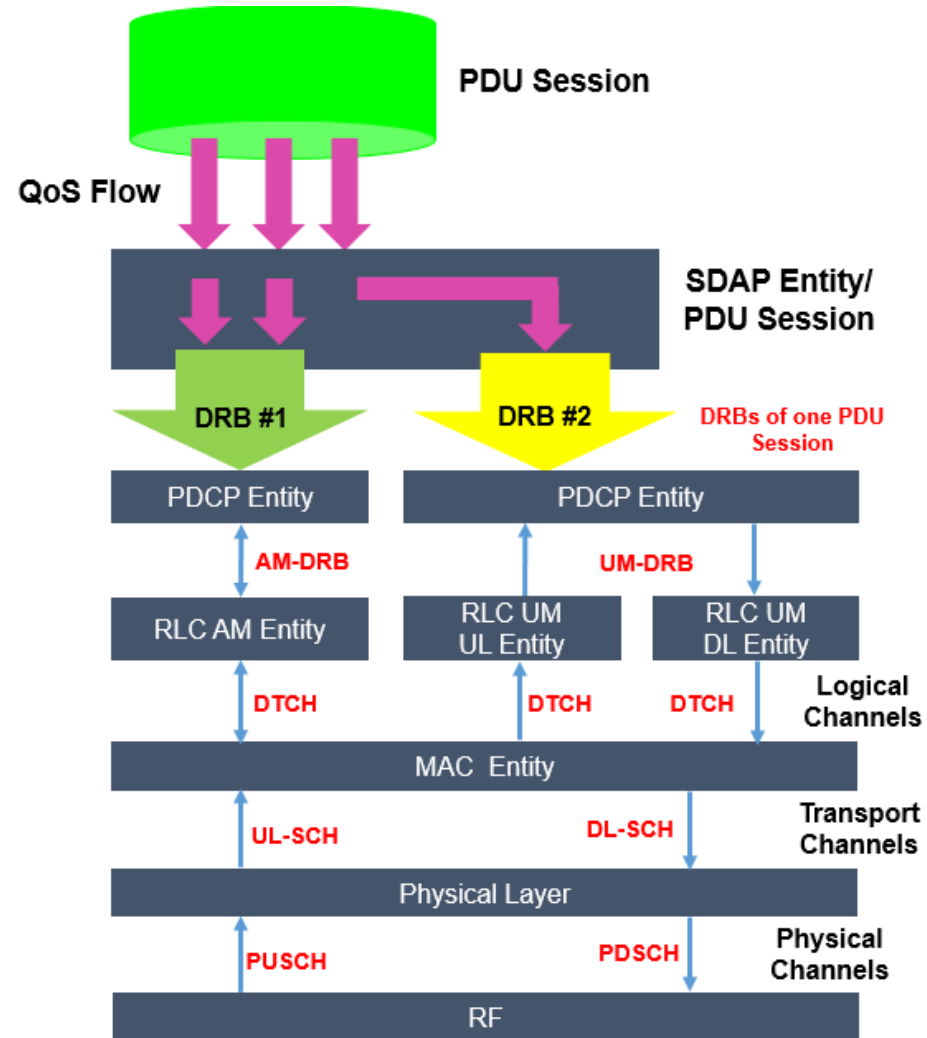
- 5G QoS Flow/SDAP
- QCI(4G) → 5QI(5G):
- * delay critical GBR ← additional GBR with Packet delay budget, Packet delay (intelligent transport, electricity distribution, discrete automation)
- * Low latency eMBB (AR-augmented Reality)
- * High-speed trains
- TS 23.501
- + new 5G layer: SDAP (with Reflective QoS)
- IoT

reflective quality of service let's talk about it a

Important pages



Important pages



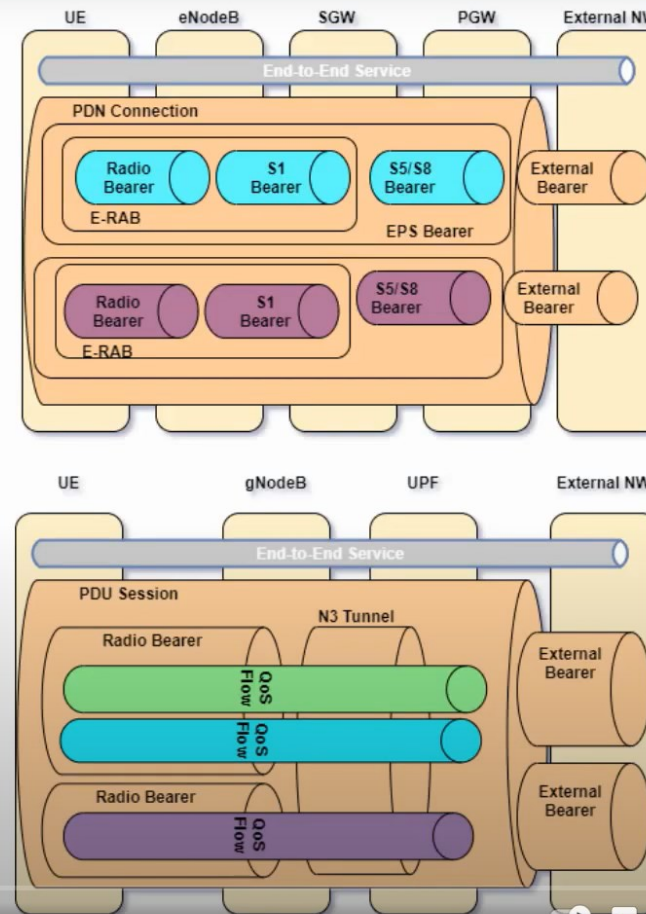
Important pages

<https://youtu.be/GUNQjwjEwQM?si=J5WmMfoxtglb8-Lf>

QoS (Quality of Service) in 5G

5G QoS

- In 5G QoS framework is based on QoS Flows which is the finest granularity of QoS differentiation.
- Each QoS Flow is identified by a QoS Flow ID.
- NG-RAN can map multiple QoS Flow IDs to one Radio Bearer or can have a 1:1 relationship. It depends on NG-RAN implementation.
- In 4G we had a 1:1 relationship between QCI and EPS bearers.
- 3GPP has standardized QoS Flow IDs like QCI for various services. More on this later.
- Also notice there is one N3 tunnel between eNB and UPF unlike multiple S5/S8 bearers in LTE.



Important pages

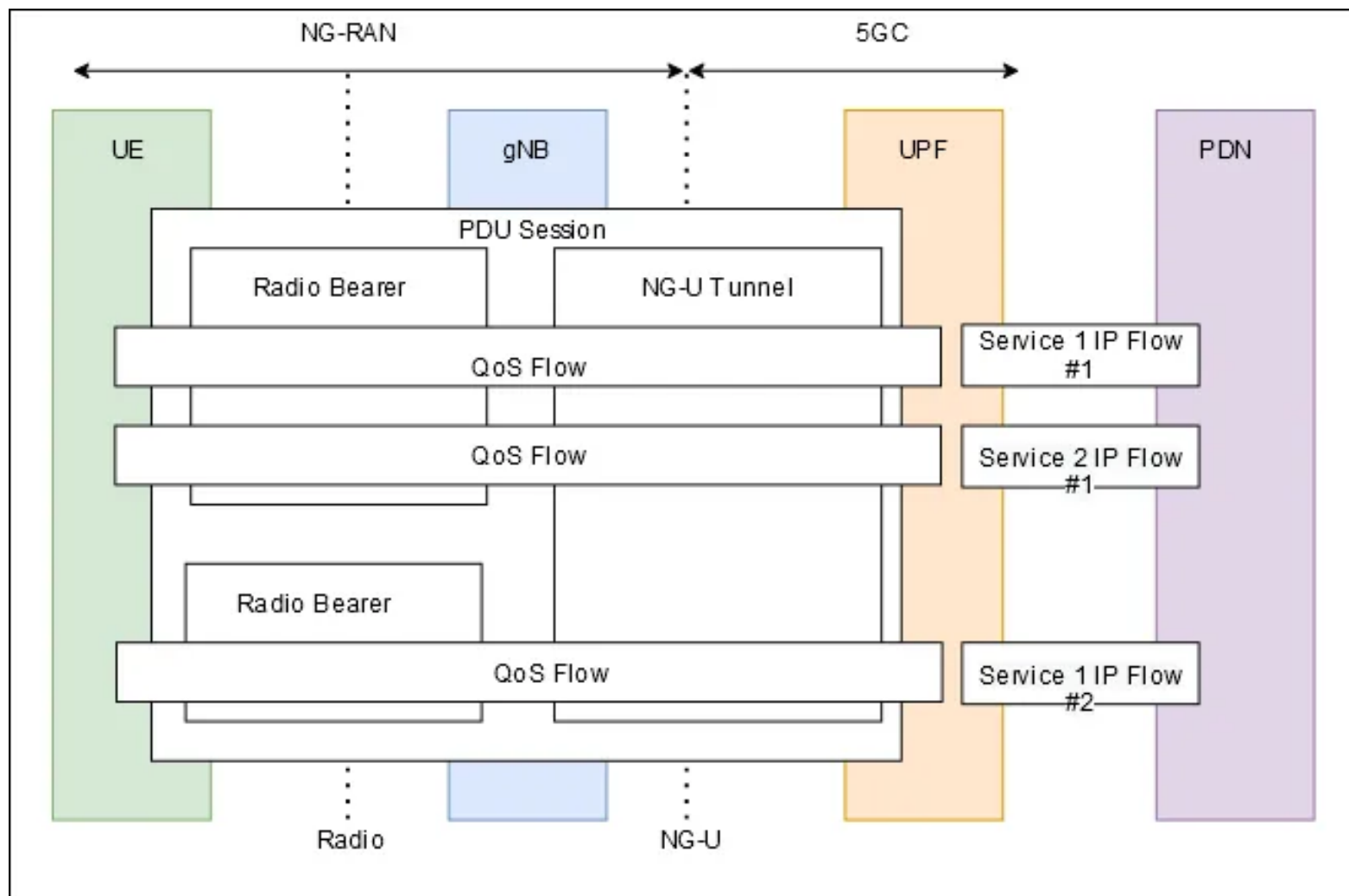
https://mp.weixin.qq.com/s/3fppz-mq_8LZZ_7IJlPhw

5G QoS与4G QoS有何不同？

Parameter	5G	4G
QoS Identifier	5G QI (QoS Identifier)	QCI (QoS Class Identifier)
IP Data Flow	QoS Flow	EPC Bearer
Flow/Bearer ID	QFI (QoS Flow Identifier)	EBI
Reflective QoS	RQI (Reflective QoS ID)	NA
Data Session	PDU Session	PDN Connection

Important pages

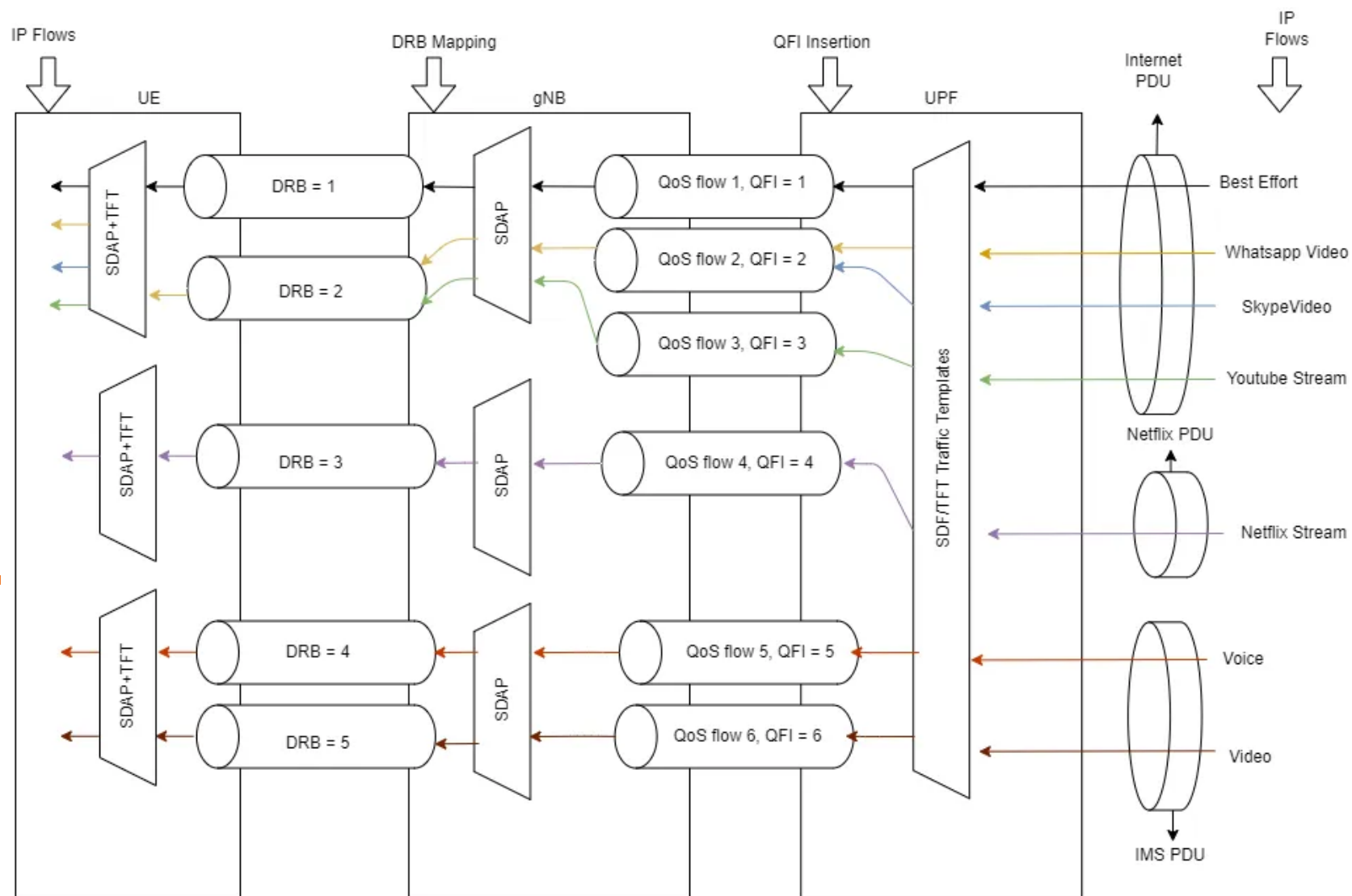
5G QoS架构



5G QoS示例

Important
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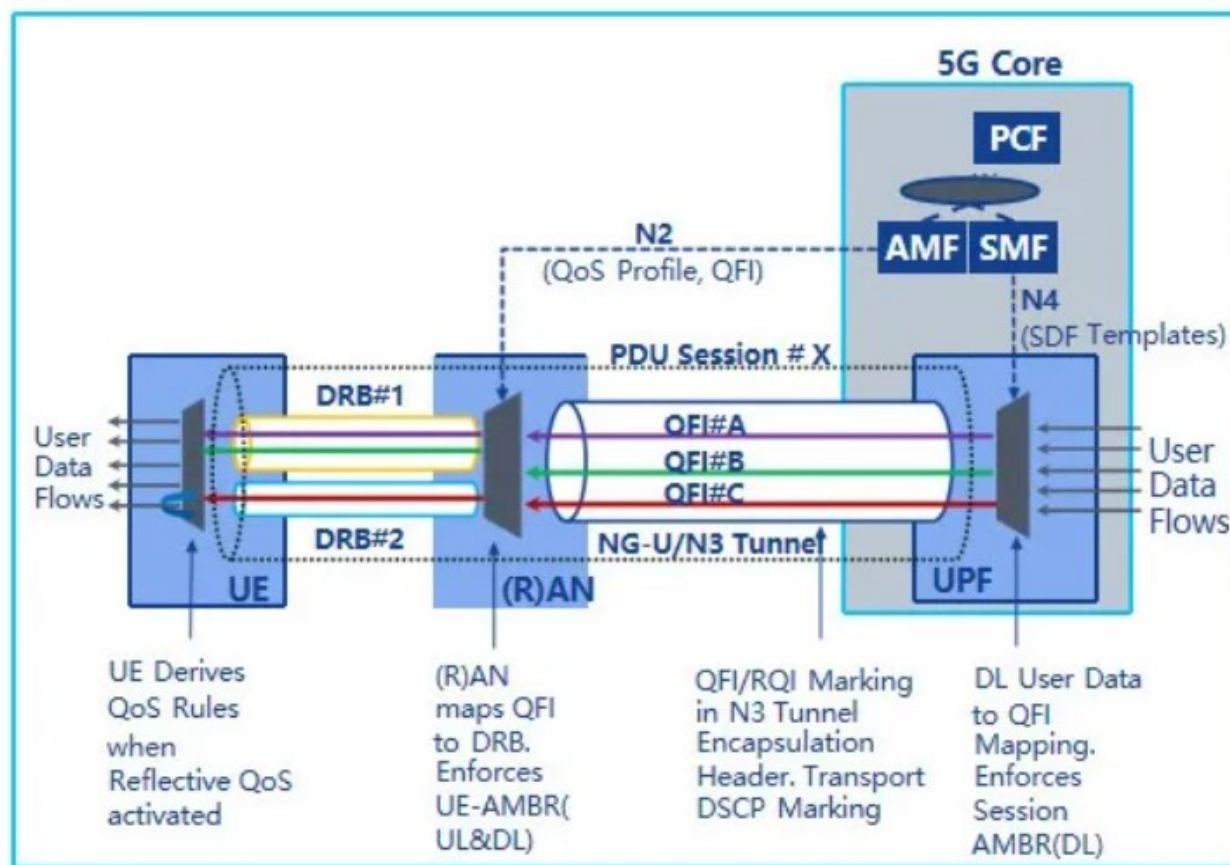
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Important pages

下行QoS流处理

<https://mp.weixin.qq.com/s/SLsT878qan6qXwQiwELR6g>



UPF将服务数据流（SDF）映射到QoS流

UPF执行Session AMBR并且还可以对PDU计数以支持计费

UPF在5GC和(R)AN之间的单个隧道中传输PDU会话的数据包，UPF封装报头中的用户平面标记（包括用于非3GPP接入的5QI）。另外，UPF可以在封装报头中包括对反射式QoS激活的指示。

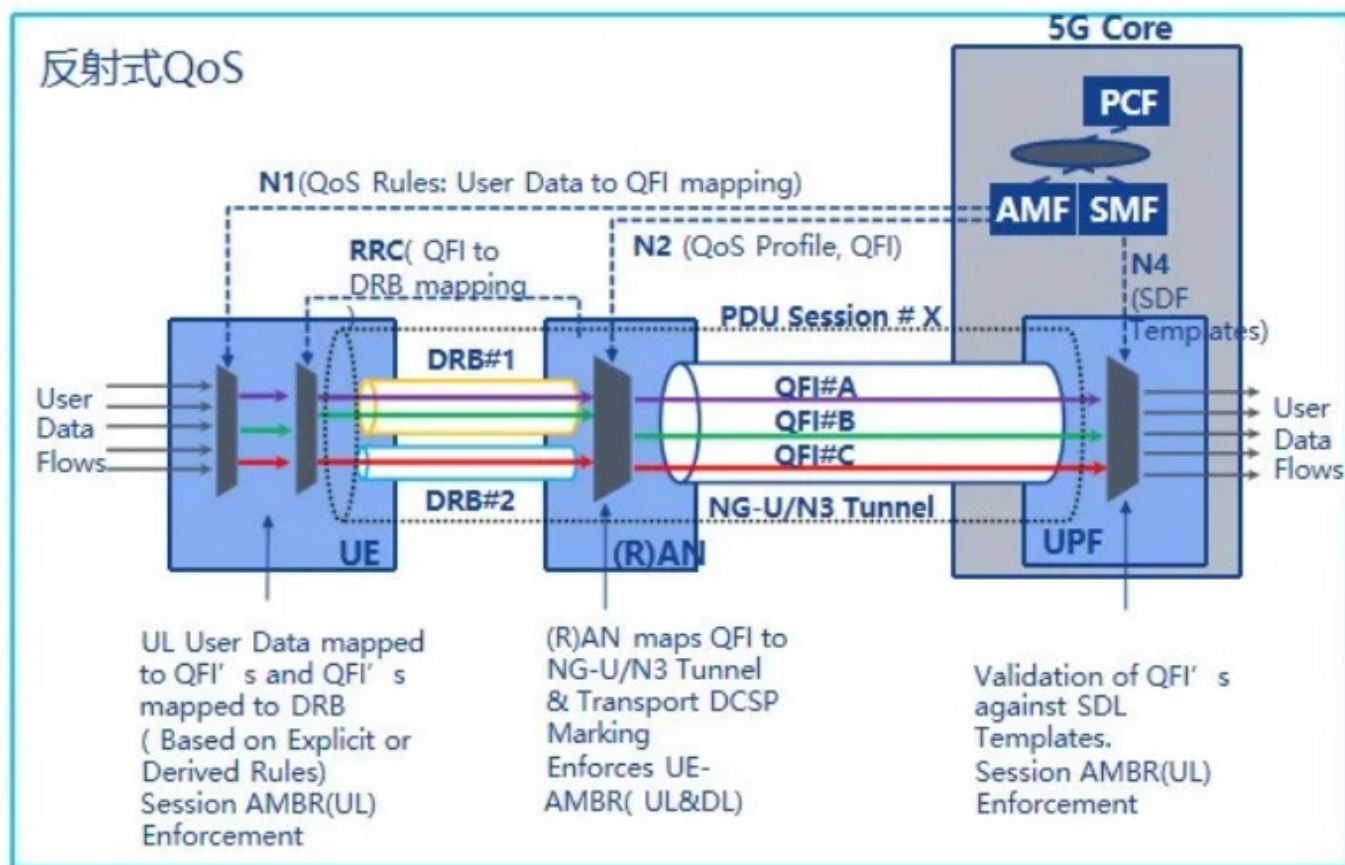
(R)AN根据QFI和相关联的5G QoS特性和参数将来自QoS流的PDU映射到特定接入的资源。

如果应用反射QoS，则UE创建新的派生QoS规则。派生QoS规则中的分组过滤器来自下行分组包（即，报头），并且派生的QoS规则的用户面标记来自于下行分组包的用户面标记。

Important pages

上行QoS流处理

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UE使用存储的QoS规则来确定SDF与QoS流之间的映射。UE基于RAN提供的映射,使用与QoS流相对应的接入特定资源来传输上行PDU。

(R)AN经由N3隧道向UPF传输PDU。在从(R)AN向CN传递上行分组包时,(R)AN可确定QFI值(该值包含在UL PDU的封装报头中),并选择N3隧道。

UPF验证UL PDU中的QFI是否与提供给UE或由UE隐式派生(例如,在采用反射式QoS的情况下)的QoS规则相一致。

UPF执行session AMBR并对PDU计数以进行计费

What We Learn here

1	Most materials have similar contents	
2	Too many keypoints	
3	WeChat has good materials	
4	UL is very different from DL	

